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Joint resource map design of DM-RS and PT-RS

Abstract

A method, network node and wireless device in a wireless communication system for one of transmitting and receiving a phase-tracking reference signal, PT-RS. The method includes obtaining information about a position in a time domain of a scheduled first demodulation reference signal, DM-RS, in a slot, and one of transmitting and receiving the PT-RS within the slot, the position of the PT-RS depending on the position in the time domain of the scheduled first DM-RS.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 18/191,618 filed Mar. 28, 2023, which itself is a continuation of U.S. application Ser. No. 16/073,680 filed Jul. 27, 2018, which is a National Stage Application of International Application No. PCT/SE2018/050629, filed Jun. 15, 2018, which claims priority to U.S. Provisional Application No. 62/521,078, filed Jun. 16, 2017, the entireties of all of which are incorporated herein by reference.

TECHNICAL FIELD

(1) This disclosure relates to wireless communications, and in particular to a method, network node and wireless device for scheduling phase tracking reference signals, PT-RS, jointly with

demodulation reference signals, DM-RS.

BACKGROUND

(2) The physical layer of New Radio (NR) (the third generation partnership project (3GPP) fifth generation (5G) mobile radio systems) is expected to handle a vast number of different transmission scenarios by operating in the frequency range from below 1 GHz to 100 GHz. Carrier frequencies above 6 GHz are not supported by long term evolution (LTE), so NR requires a new and flexible design for the physical layer which offers good performance in a wider range of frequencies than the physical layer of LTE.

(3) Similar to LTE, NR will use orthogonal frequency division multiplexing (OFDM) based waveforms with reference signals and physical layer channels mapped on a time-frequency resource grid. NR has an ultra-lean design that minimizes always-on transmissions to enhance network energy efficiency and ensure forward compatibility. In contrast to the setup in LTE, the reference signals in NR are transmitted only when necessary. Demodulation reference signals (DM-RS) and phase-tracking reference signals (PT-RS) are two variations of reference signals among the others.

(4) DM-RS is used to estimate the radio channel for demodulation. DM-RS is wireless device-specific, can be beamformed, confined in a scheduled resource, and transmitted only when necessary, both in downlink (DL), i.e., from base station to wireless device, and uplink (UL), i.e., from wireless device to base station. To support multiple-layer multiple input multiple output (MIMO) transmission, multiple orthogonal DM-RS ports can be scheduled, one for each layer. Orthogonality is achieved by frequency division multiplexing (FDM) (comb structure), time division multiplexing (TDM) and code division multiplexing (CDM) (with cyclic shift of the root sequence or orthogonal cover codes). The basic DM-RS pattern is front loaded, as the DM-RS design takes into account the early decoding requirement to support low-latency applications. For low-speed scenarios, DM-RS uses low density in the time domain. However, for high-speed scenarios, the time density of DM-RS is increased to track fast changes in the radio channel. FIGS. 1 and 2 illustrate the potential DM-RS resource mapping in the frequency-time grid for low Doppler and high Doppler scenarios, respectively, in the transmission slot.

(5) Another challenge that NR faces is the radio-frequency (RF) impairments when wireless systems operate in the millimeter (mm) wave band, specifically, the effects of phase noise produced by the local oscillators. The degradation produced by phase noise increases as the carrier frequency increases, so that the physical layer of NR operating in mmWave frequencies has to be robust to phase noise in order to achieve good performance. Hence, there is a need for a new reference signal called the phase tracking reference signal (PT-RS). Such signal can be used both for mitigation of the phase noise-induced common phase error (CPE), experienced equally on all subcarriers within an OFDM symbol, and inter-carrier interference (ICI) caused by the loss of orthogonality between subcarriers.

(6) The PT-RS may be needed both in uplink and downlink. It is foreseen that this signal can be used for both fine carrier frequency-synchronization and phase noise compensation. This signal is assumed to be present and needed only at high carrier frequencies, while the other properties of the DM-RS can remain somewhat unchanged. An example of adding a PT-RS at high carrier frequencies is depicted in FIG. 3.

(7) Different modulation and coding schemes (MCS) offer different robustness against the effects of phase noise, as shown in FIGS. 4 and 5. Therefore, the time density of PT-RS for a specific wireless device (WD) can be configured according to the scheduled MCS.

(8) The problems of the existing solution can be summarized as follows: PT-RS is a new reference-signal introduced to NR, which may co-exist with DM-RS in certain scenarios; Each type of reference signal needs to reserve its own resource in the time-frequency grid; Prior art solutions propose a joint design for DM-RS and PT-RS but the design is limited to the case in which just one DM-RS instance is scheduled in the slot; and Pilot contamination may happen such that the

overhead of the total resources used by reference signals needs to be controlled.

SUMMARY

(9) The proposed solution for the joint design of DM-RS and PT-RS may be based on the conditions that the PT-RS mapping in the time domain may depend at least on the following: The position of the front-loaded DM-RS in the slot for an antenna port in which PT-RS is mapped; Required time density in the time domain for PT-RS; The position of the first symbol scheduled for data transmission in the transmission slot; and The position of the last symbol scheduled for data transmission in the transmission slot.

The proposed solution may also include joint mapping of DM-RS and PT-RS. The position of the additional DM-RS in the slot may depend on the PT-RS which is mapped.

(10) One aspect is that the PT-RS resource can be aligned with the first DM-RS in the slot when PT-RS is scheduled. Furthermore, when the additional DM-RS and PT-RS both exist in the resource grid, the additional DM-RS position may align with the PT-RS position.

(11) In some embodiments, a method for use in a radio node in a wireless communication system for one of transmitting and receiving a phase tracking-reference signal (PT-RS) is provided. The method includes obtaining information about a position in a time domain of a scheduled first demodulation reference signal (DM-RS) in a slot. The method also includes one of transmitting and receiving the PT-RS within the slot, where the position of the PT-RS depends on the position in the time domain of the scheduled first DM-RS.

(12) In some embodiments, the obtaining comprises one of receiving information about and determining the position in the time domain of the scheduled first DM-RS. In some embodiments, the method further includes obtaining information about a position of a first time symbol in a slot scheduled for data transmission. In some embodiments, the method also includes obtaining information about a position of a last time symbol in the slot scheduled for data transmission. In some embodiments, the method also includes obtaining information indicating a scheduled modulation and coding scheme, MCS, and transmitting the PT-RS with a time density based on the scheduled MCS. In some embodiments, the time density is one 1, $\frac{1}{2}$ and $\frac{1}{4}$. In some embodiments, the method includes mapping the PT-RS to resource elements, REs, in the slot based on one or more of a position of the scheduled first DM-RS, a scheduled MCS, a required time-density, a position of the first time symbol scheduled for data transmission and a position of the last time symbol scheduled for data transmission. In some embodiments, the first DM-RS is scheduled in resource elements, REs, which span several subcarriers in frequency and at least one time symbol of the slot in time, while the PT-RS is one of transmitted and received in REs which span at least one subcarrier in frequency and multiple time symbols of the slot in time. In some embodiments, a physical resource block, PRB, of the slot has 12 subcarriers in the frequency domain and one of 12 and 14 time symbols in the time domain. In some embodiments, the radio node is one of a WD and a network node.

(13) According to another aspect, a radio node in a wireless communication system configured for one of transmitting and receiving a PT-RS is provided. The radio node includes processing circuitry configured to obtain information about a position in a time domain of a scheduled first DM-RS in a slot. The processing circuitry is further configured to one of transmit and receive the PT-RS within the slot, where the position of the PT-RS depends on the position in the time domain of the scheduled first DM-RS.

(14) According to this aspect, in some embodiments, the obtaining comprises one of receiving information about and determining the position in the time domain of the scheduled first DM-RS. In some embodiments, the processing circuitry is further configured to obtain information about a position of a first time symbol in a slot scheduled for data transmission, and obtain information about a position of a last time symbol in the slot scheduled for data transmission. In some embodiments, the processing is further configured to obtain information indicating a scheduled modulation and coding scheme, MCS, and transmit the PT-RS with a time density based on the

scheduled MCS. In some embodiments, the time density is one 1, $\frac{1}{2}$ and $\frac{1}{4}$. In some embodiments, the processing circuitry is further configured to map the PT-RS to resource elements, REs, in the slot based on one or more of a position of the scheduled first DM-RS, a scheduled MCS, a required time-density, a position of the first time symbol scheduled for data transmission and a position of the last time symbol scheduled for data transmission. In some embodiments, the first DM-RS is scheduled in resource elements, REs, which span several subcarriers in frequency and at least one time symbol of the slot in time, while the PT-RS is one of transmitted and received in REs which span at least one subcarrier in frequency and multiple time symbols of the slot in time. In some embodiments, a physical resource block, PRB, of the slot has 12 subcarriers in the frequency domain and one of 12 and 14 time symbols in the time domain. In some embodiments, the radio node is one of a WD and a network node.

(15) According to another aspect, a radio node in a wireless communication system configured for one of transmitting and receiving a PT-RS is provided. The radio node includes demodulation reference signal, DM-RS, position module configured to obtain information about a position in a time domain of a scheduled first DM-RS in a slot. The radio node further includes a PT-RS transceiver module configured to one of transmit and receive the PT-RS within the slot, where the position of the PT-RS depends on the position in the time domain of the scheduled first DM-RS. In some embodiments, the first DM-RS is scheduled in resource elements, REs, which span several subcarriers in frequency and at least one single time symbol of the slot in time, while the PT-RS is one of transmitted and received in REs which span at least one subcarrier in frequency and multiple time symbols of the slot in time.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A more complete understanding of embodiments described herein, and the attendant advantages and features thereof, will be more readily understood by reference to the following detailed description when considered in conjunction with the accompanying drawings wherein:
- (2) FIG. 1 shows possible DM-RS patterns for NR that support early decoding for low Doppler;
- (3) FIG. 2 shows possible DM-RS patterns for NR that support early decoding for high Doppler;
- (4) FIG. 3 shows addition of PT-RS at high carrier frequencies;
- (5) FIG. 4 shows evaluation results for 16 QAM ($\frac{3}{4}$) MCS;
- (6) FIG. 5 shows evaluation results for 64 QAM ($\frac{5}{6}$) MCS;
- (7) FIG. 6 is a block diagram of a wireless communication system constructed in accordance with principles set forth herein;
- (8) FIG. 7 is a block diagram of a network node constructed in accordance with principles set forth herein;
- (9) FIG. 8 is a block diagram of an alternative embodiment of a network node constructed in accordance with principles set forth herein;
- (10) FIG. 9 is a block diagram of a wireless device constructed in accordance with principles set forth herein;
- (11) FIG. 10 is a block diagram of an alternative embodiment of a wireless device constructed in accordance with principles set forth herein; and
- (12) FIG. 11 is a flowchart of an exemplary process for joint scheduling of DM-RS and PT-RS that can be performed in the wireless device and/or the network node.
- (13) FIG. 12 is a flowchart of an exemplary process for scheduling DM-RS and PT-RS;
- (14) FIG. 13 is a flowchart of an exemplary process for transmitting and receiving a phase tracking reference signal (PT-RS);
- (15) FIG. 14 shows a single DM-RS instance with aligned DM-RS and PT-RS with time density 1;

(16) FIG. 15 shows a single front loaded DM-RS with aligned DM-RS and PT-RS with time density $\frac{1}{2}$;

(17) FIG. 16 shows a double front loaded DM-RS with aligned DM-RS and PT-RS with time density $\frac{1}{2}$;

(18) FIG. 17 shows front loaded DM-RS pattern with additional DM-RS and PT-RS with time density 1;

(19) FIG. 18 shows front loaded DM-RS pattern with additional DM-RS and PT-RS with time density $\frac{1}{2}$;

DETAILED DESCRIPTION

(20) Before describing in detail exemplary embodiments, it is noted that the embodiments reside primarily in combinations of apparatus components and processing steps related to scheduling phase tracking reference signals, PT-RS, jointly with demodulation reference signals, DM-RS. Accordingly, the system and method components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

(21) As used herein, relational terms, such as “first” and “second,” “top” and “bottom,” and the like, may be used solely to distinguish one entity or element from another entity or element without necessarily requiring or implying any physical or logical relationship or order between such entities or elements.

(22) An indication generally may explicitly and/or implicitly indicate the information it represents and/or indicates. Implicit indication may for example be based on position and/or resource used for transmission. Explicit indication may for example be based on a parametrization with one or more parameters, and/or one or more index or indices, and/or one or more bit patterns representing the information. It may in particular be considered that control signaling as described herein, based on the utilized resource sequence, implicitly indicates the control signaling type.

(23) The term signal used herein can be any physical signal or physical channel. Examples of physical signals are reference signal such as PSS, SSS, CRS, PRS etc. The term physical channel (e.g., in the context of channel reception) used herein is also called as ‘channel. Examples of physical channels are MIB, PBCH, NPBCH, PDCCH, PDSCH, sPUCCH, SPDSCH, sPUCCH, sPUSCH, MPDCCH, NPDCCH, NPDSCH, E-PDCCH, PUSCH, PUCCH, NPUSCH etc. These terms/abbreviations may be used according to 3GPP standard language, in particular according to LTE and/or NR.

(24) It may be considered for cellular communication there is provided at least one uplink (UL) connection and/or channel and/or carrier and at least one downlink (DL) connection and/or channel and/or carrier, e.g., via and/or defining a cell, which may be provided by a network node, in particular a base station or eNodeB. An uplink direction may refer to a data transfer direction from a terminal to a network node, e.g., base station and/or relay station. A downlink direction may refer to a data transfer direction from a network node, e.g., base station and/or relay node, to a terminal. UL and DL may be associated to different frequency resources, e.g., carriers and/or spectral bands. A cell may comprise at least one uplink carrier and at least one downlink carrier, which may have different frequency bands. A network node, e.g., a base station, gNB or eNodeB, may be adapted to provide and/or define and/or control one or more cells, e.g., a PCell and/or a LA cell.

(25) Transmitting in downlink may pertain to transmission from the network or network node to the terminal. Transmitting in uplink may pertain to transmission from the terminal to the network or network node. Transmitting in sidelink may pertain to (direct) transmission from one terminal to another. Uplink, downlink and sidelink (e.g., sidelink transmission and reception) may be considered communication directions. In some variants, uplink and downlink may also be used to described wireless communication between network nodes, e.g. for wireless backhaul and/or relay

communication and/or (wireless) network communication for example between base stations or similar network nodes, in particular communication terminating at such. It may be considered that backhaul and/or relay communication and/or network communication is implemented as a form of sidelink or uplink communication or similar thereto.

(26) Generally, configuring may include determining configuration data representing the configuration and providing, e.g. transmitting, it to one or more other nodes (parallel and/or sequentially), which may transmit it further to the radio node (or another node, which may be repeated until it reaches the wireless device). Alternatively, or additionally, configuring a radio node, e.g., by a network node or other device, may include receiving configuration data and/or data pertaining to configuration data, e.g., from another node like a network node **16**, which may be a higher-level node of the network, and/or transmitting received configuration data to the radio node. Accordingly, determining a configuration and transmitting the configuration data to the radio node may be performed by different network nodes or entities, which may be able to communicate via a suitable interface, e.g., an X2 interface in the case of LTE or a corresponding interface for NR. Configuring a terminal (e.g. WD) may comprise scheduling downlink and/or uplink transmissions for the terminal, e.g. downlink data and/or downlink control signaling and/or DCI and/or uplink control or data or communication signaling, in particular acknowledgement signaling, and/or configuring resources and/or a resource pool therefor. In particular, configuring a terminal (e.g. WD) may comprise configuring the WD to perform certain measurements on certain subframes or radio resources and reporting such measurements according to embodiments of the present disclosure

(27) Signaling may comprise one or more signals and/or symbols. Reference signaling may comprise one or more reference signals and/or symbols. Data signaling may pertain to signals and/or symbols containing data, in particular user data and/or payload data and/or data from a communication layer above the radio and/or physical layer/s. It may be considered that demodulation reference signaling comprises one or more demodulation signals and/or symbols. Demodulation reference signaling may in particular comprise DM-RS according to 3GPP and/or NR and/or LTE technologies. Demodulation reference signaling may generally be considered to represent signaling providing reference for a receiving device like a terminal to decode and/or demodulate associated data signaling or data. Demodulation reference signaling may be associated to data or data signaling, in particular to specific data or data signaling. It may be considered that data signaling and demodulation reference signaling are interlaced and/or multiplexed, e.g. arranged in the same time interval covering e.g. a subframe or slot or symbol, and/or in the same time-frequency resource structure like a resource block. A resource element may represent a smallest time-frequency resource, e.g. representing the time and frequency range covered by one symbol or a number of bits represented in a common modulation. A resource element may e.g. cover a symbol time length and a subcarrier, in particular in 3GPP and/or NR and/or LTE standards. A data transmission may represent and/or pertain to transmission of specific data, e.g. a specific block of data and/or transport block. Generally, demodulation reference signaling may comprise and/or represent a sequence of signals and/or symbols, which may identify and/or define the demodulation reference signaling.

(28) Data or information may refer to any kind of data, in particular any one of and/or any combination of control data or user data or payload data. Control information (which may also be referred to as control data) may refer to data controlling and/or scheduling and/or pertaining to the process of data transmission and/or the network or terminal operation.

(29) FIG. **6** is a block diagram of a wireless communication network configured according to principles set forth herein. The wireless communication network **10** includes a cloud **12** which may include the Internet and/or the public switched telephone network (PSTN). Cloud **12** may also serve as a backhaul network of the wireless communication network **10**. The wireless communication network **10** includes one or more network nodes **14A** and **14B**, which may

communicate directly via an X2 interface in LTE embodiments, and are referred to collectively as network nodes **14**. It is contemplated that other interface types can be used for communication between network nodes **14** for other communication protocols such as New Radio (NR). The network nodes **14** may serve wireless devices **16A** and **16B**, referred to collectively herein as wireless devices **16**. Note that, although only two wireless devices **16** and two network nodes **14** are shown for convenience, the wireless communication network **10** may typically include many more wireless devices (WDs) **16** and network nodes **14**. Further, in some embodiments, WDs **16** may communicate directly using what is sometimes referred to as a side link connection.

(30) The term “wireless device” or mobile terminal used herein may refer to any type of wireless device communicating with a network node **14** and/or with another wireless device **16** in a cellular or mobile communication system **10**. Examples of a wireless device **16** are user equipment (UE), target device, device to device (D2D) wireless device, machine type wireless device or wireless device capable of machine to machine (M2M) communication, PDA, tablet, smart phone, laptop embedded equipped (LEE), laptop mounted equipment (LME), USB dongle, etc.

(31) The term “network node” used herein may refer to any kind of radio base station in a radio network which may further comprise any base transceiver station (BTS), base station controller (BSC), radio network controller (RNC), evolved Node B (eNB or eNodeB), NR gNodeB, NR gNB, Node B, multi-standard radio (MSR) radio node such as MSR BS, relay node, donor node controlling relay, radio access point (AP), transmission points, transmission nodes, Remote Radio Unit (RRU) Remote Radio Head (RRH), nodes in distributed antenna system (DAS), etc.

(32) Although embodiments are described herein with reference to certain functions being performed by network node **14**, it is understood that the functions can be performed in other network nodes and elements. It is also understood that the functions of the network node **14** can be distributed across network cloud **12** so that other nodes can perform one or more functions or even parts of functions described herein. Also, functions described herein as being performed by a network node **14** may also be performed by a wireless device **16**.

(33) The network node **14** has DM-RS position information **30** that may be stored in memory. The DM-RS position information includes information about a position in the time domain of a scheduled first DM-RS in a slot. The network node **14** also has a PT-RS transceiver **28** configured to transmit or receive a PT-RS within the slot, where the position of the PT-RS depends on a position of the DM-RS. Similarly, the wireless device **16** may include DM-RS position information **50** and a PT-RS transceiver **48** that perform the same functions as the DM-RS position memory **30** and the PT-RS transceiver **28**, respectively.

(34) FIG. 7 is a block diagram of a network node **14** configured for joint scheduling of DM-RS and PT-RS. The network node **14** has processing circuitry **22**. In some embodiments, the processing circuitry may include a memory **24** and processor **26**, the memory **24** containing instructions which, when executed by the processor **26**, configure processor **26** to perform the one or more functions described herein. In addition to a traditional processor and memory, processing circuitry **22** may include integrated circuitry for processing and/or control, e.g., one or more processors and/or processor cores and/or FPGAs (Field Programmable Gate Array) and/or ASICs (Application Specific Integrated Circuitry).

(35) Processing circuitry **22** may include and/or be connected to and/or be configured for accessing (e.g., writing to and/or reading from) memory **24**, which may include any kind of volatile and/or non-volatile memory, e.g., cache and/or buffer memory and/or RAM (Random Access Memory) and/or ROM (Read-Only Memory) and/or optical memory and/or EPROM (Erasable Programmable Read-Only Memory). Such memory **24** may be configured to store code executable by control circuitry and/or other data, e.g., data pertaining to communication, e.g., configuration and/or address data of nodes, etc. Processing circuitry **22** may be configured to control any of the methods described herein and/or to cause such methods to be performed, e.g., by processor **26**. Corresponding instructions may be stored in the memory **24**, which may be readable and/or

readably connected to the processing circuitry **22**. In other words, processing circuitry **22** may include a controller, which may include a microprocessor and/or microcontroller and/or FPGA (Field-Programmable Gate Array) device and/or ASIC (Application Specific Integrated Circuit) device. It may be considered that processing circuitry **22** includes or may be connected or connectable to memory, which may be configured to be accessible for reading and/or writing by the controller and/or processing circuitry **22**.

(36) The memory **24** is configured to store DM-RS position information **30** and PT-RS schedule information **32**. The processor **26** implements a DM-RS position obtainer **18** configured to obtain information about a position in the time domain of a scheduled first DM-RS in a slot. The processor **26** may also implement a PT-RS scheduler **20** configured to schedule the PT-RS in the slot. A transceiver **28** is configured to transmit the PT-RS to a wireless device **16** or receive the PT-RS from the wireless device **16**, where the position of the PT-RS depends on the position of the first DM-RS.

(37) FIG. **8** is a block diagram of an alternative embodiment of a network node **14** configured for joint scheduling of DM-RS and PT-RS. The memory module **25** is configured to store DM-RS position information **30** and PT-RS schedule information **32**. The DM-RS position obtainer module **19** is configured to obtain information about a position in the time domain of a scheduled first DM-RS in a slot. The PT-RS scheduler module **21** is configured to schedule the PT-RS in the slot. A transceiver module **29** is configured to transmit the PT-RS to a wireless device **16** or receive a PT-RS from the wireless device **16**, where the position of the PT-RS depends on the position of the first DM-RS.

(38) Note that the same components shown in FIG. **7** in the network node **14** can be implemented in a wireless device **16** for joint scheduling of DM-RS and PT-RS by the wireless device **16** for transmission on the uplink. Thus, the wireless device **16** may have a DM-RS position obtainer **58** and a PT-RS scheduler **60** for joint scheduling of the DM-RS and PT-RS on the uplink.

(39) Accordingly, FIG. **9** is a block diagram of a wireless device **16** configured for joint scheduling of DM-RS and PT-RS. The wireless device **16** has processing circuitry **42**. In some embodiments, the processing circuitry may include a memory **44** and processor **46**, the memory **44** containing instructions which, when executed by the processor **46**, configure processor **46** to perform the one or more functions described herein. In addition to a traditional processor and memory, processing circuitry **42** may include integrated circuitry for processing and/or control, e.g., one or more processors and/or processor cores and/or FPGAs (Field Programmable Gate Array) and/or ASICs (Application Specific Integrated Circuitry).

(40) Processing circuitry **42** may include and/or be connected to and/or be configured for accessing (e.g., writing to and/or reading from) memory **44**, which may include any kind of volatile and/or non-volatile memory, e.g., cache and/or buffer memory and/or RAM (Random Access Memory) and/or ROM (Read-Only Memory) and/or optical memory and/or EPROM (Erasable Programmable Read-Only Memory). Such memory **44** may be configured to store code executable by control circuitry and/or other data, e.g., data pertaining to communication, e.g., configuration and/or address data of nodes, etc. Processing circuitry **42** may be configured to control any of the methods described herein and/or to cause such methods to be performed, e.g., by processor **46**. Corresponding instructions may be stored in the memory **44**, which may be readable and/or readably connected to the processing circuitry **42**. In other words, processing circuitry **42** may include a controller, which may include a microprocessor and/or microcontroller and/or FPGA (Field-Programmable Gate Array) device and/or ASIC (Application Specific Integrated Circuit) device. It may be considered that processing circuitry **42** includes or may be connected or connectable to memory, which may be configured to be accessible for reading and/or writing by the controller and/or processing circuitry **42**.

(41) The memory **44** is configured to store DM-RS position information **50** and PT-RS schedule information **52**. The processor **46** implements a DM-RS position obtainer **58** configured to obtain

information about a position in the time domain of a scheduled first DM-RS in a slot. The processor **46** also implements a PT-RS scheduler **60** configured to schedule the PT-RS in the slot. A transceiver **48** is configured to transmit the PT-RS to a network node **14** or receive a PT-RS from the network node **14**, where the position of the PT-RS depends on the position in the time domain of the scheduled first DM-RS.

(42) FIG. **10** is a block diagram of an alternative embodiment of a wireless device **16** configured for joint scheduling of DM-RS and PT-RS. The memory module **45** is configured to store DM-RS position information **50** and PT-RS schedule information **52**. The DM-RS position obtainer module **59** is configured to obtain information about a position in the time domain of a scheduled first DM-RS in a slot. The PT-RS scheduler module **61** is configured to schedule the PT-RS in the slot. A transceiver module **49** is configured to transmit the PT-RS to a network node **14** or receive a PT-RS from the network node **14**, where the position of the PT-RS depends on the position in the time domain of the scheduled first DM-RS.

(43) FIG. **11** is a flowchart of an exemplary process for joint scheduling of DM-RS and PTRS. This process can be performed in the network node **14** and/or in the wireless device **16**. The process includes scheduling, via the processor **26**, **46**, the DM-RS in a time-frequency resource grid at a plurality of frequencies in a same time slot (block **S90**). The process also includes scheduling, via the PT-RS scheduler **20**, **60**, the PT-RS in the time-frequency grid at a plurality of time slots at a same frequency, the time slots at which the PT-RS are scheduled depending on a position of the DM-RS (block **S92**).

(44) An advantage of some embodiments is that the total reference signal overhead can be reduced to avoid pilot contamination, while achieving the required estimation quality. The main steps of an embodiment of the proposed joint design of DM-RS and PT-RS position are shown in FIG. **12**. The process includes determining, via the PT-RS scheduler **20**, **60** a PT-RS time density according to a scheduled MCS (block **S100**). The process also includes determining, via the DM-RS position obtainer, **18**, **58**, a position of a front loaded DM-RS (block **S102**). The process further includes determining, via the processor **26**, **46**, a position of the first and last symbol scheduled for data transmission (block **S104**). Then the solution proposed and described herein is used to obtain a map of the PT-RS in a slot (block **S106**). If there are additional instances of the DM-RS (block **S108**), then the proposed solution is used to obtain the DM-RS map (block **S110**). Otherwise, the process concludes.

(45) FIG. **13** is a flowchart of an exemplary process for transmitting or receiving a phase tracking reference signal (PT-RS). The process includes obtaining, via the DM-RS position obtainer **18**, **48**, information about a position in a time domain of a scheduled first demodulation reference signal, DM-RS in a slot (block **S120**). The position may be obtained from the WD **16** via radio resource control scheduling of the DM-RS. The process also includes one of transmitting and receiving, via the PT-RS transceiver **28**, **48**, the PT-RS within the slot, where the position of the PT-RS depends on the position in the time domain of the scheduled first DM-RS (block **S122**).

(46) Having described the general process flow of arrangements of the disclosure and having provided examples of hardware and software arrangements for implementing the processes and functions of the disclosure, the sections below provide details and examples of arrangements for implementing embodiments of the disclosure and for transmitting and receiving PT-RS.

(47) The proposed solution may have at least some of the following benefits: a reduction in the total overhead of the reference signals when DM-RS and PT-RS are both scheduled; a common design of both reference signals which adapts with the actual position of the front-loaded DM-RS in the slot; and both types of reference signal can provide desired estimation quality.

The process may be described as follows, in some embodiments. Consider a slot-based transmission. A slot-interval may be L OFDM symbol length, e.g., 7 or 14 symbols. The symbol index has range $[1:L]$. The mapping of the PT-RS in the time domain is determined by: $X_{\text{sub.REF}} = \min(D_{\text{sub.FL}})$, where $D_{\text{sub.FL}}$ represents the set containing the position of the front-

loaded DM-RS. $\Delta_{\text{sub.PTRS}}$, the distance between PT-RS instances, which is inversely proportional to the PT-RS time density. $D_{\text{sub}.0} \geq 1$, the first symbol scheduled for data transmission in the slot. $D_{\text{sub}.1} \leq L$, the last symbol scheduled for data transmission in the slot.

Let P represent the set containing the time position of PT-RS in the transmission slot. In order to obtain an aligned design for front-loaded DM-RS and PT-RS, P may be defined as follows:

$$P = \{n \in \text{custom character} / D_{\text{sub}.0} \leq n \leq D_{\text{sub}.1} \text{ and } (n - X_{\text{sub}.REF}) \bmod \Delta_{\text{sub.PTRS}} = 0\}$$

Let D represent the set of potential positions of the additional DM-RS in the slot. In order to obtain an aligned design for DM-RS and PT-RS, D may be defined as follows:

$$D = \{n \in P \text{ and } n > X_{\text{sub}.REF}\}$$

So the DM-RS instances are a subset of the PT-RS instances, offering an aligned design for PT-RS and DM-RS. To summarize, the overall criteria of a proposed solution according to some embodiments may be expressed as:

$$(48) P, D = \arg\max_{P, D} \text{Math. } P \cdot \text{Math. } D \cdot \text{subject to } X_{REF}, P_{TRS}, D_0, D_1.$$

Embodiment: Front Loaded DM-RS and PT-RS with Time Density 1

(49) In FIG. 14, an example of joint design for PT-RS with time density 1 and single front-loaded DMRS instance pattern is shown.

(50) In FIG. 15, an example of joint design for single front-loaded DM-RS pattern and PT-RS with time density $\frac{1}{2}$ is shown. It can be seen that by using a joint design for DM-RS and PT-RS, the reference signal overhead in the slot may be kept low (because the DM-RS instance can be re-used for phase noise estimation, replacing PT-RS). Without the joint design the reference signal time density could be higher than $\frac{1}{2}$.

(51) FIG. 16 is an example of a joint design for a double front-loaded DMRS pattern and PT-RS with time density $\frac{1}{2}$. A benefit of the proposed joint design in some embodiments is that the PT-RS map does not change for single and front-loaded DMRS pattern.

Embodiment: Front-Loaded DM-RS with Additional DM-RS Instance and PT-RS with Time Density 1

(52) In FIG. 17, an example of a joint design for front loaded DM-RS pattern with additional DM-RS and PT-RS with time density 1 is shown.

Embodiment: Front-Loaded DM-RS with Additional DM-RS Instance and PT-RS with Time Density $\frac{1}{2}$

(53) In FIG. 18, an example of a joint design for a front-loaded DMRS pattern with an additional DM-RS instance and PT-RS with time density $\frac{1}{2}$ is shown. By using a joint design for DM-RS and PT-RS, the reference signal overhead in the time domain in the slot may be kept low (because the DM-RS instances can be re-used for phase noise estimation, replacing PT-RS). Without the joint design the reference signal time density could be higher than $\frac{1}{2}$.

(54) An advantage of some embodiments is that a joint design of DM-RS and PT-RS can reduce the overhead of the reference signals in the time domain while keeping accuracy in the estimations based on the reference signals.

(55) The PT-RS is usually scheduled for only one antenna port (associated with one DMRS port), i.e., there is no multiplexing of resources used by the PT-RS, which paves the way for the alignment of DM-RS and PT-RS resource elements to reduce overhead. The density of the PT-RS depends on the modulation and coding scheme (MCS), which provides the freedom of placing the PT-RS in the resource grid using different time-offsets with equal performance. Because the PT-RS is a time domain signal which spans the resource grid with a certain density, it also provides a set of positions with which the DM-RS can align. A result is that when the additional DMRS are scheduled, there exists at least one PT-RS resource position that provides a desired channel estimation quality.

(56) In some embodiments, a method for use in a radio node 14, 16 in a wireless communication system for one of transmitting and receiving a phase tracking-reference signal, PT-RS, is provided. The method includes obtaining (block S120) information about a position in a time domain of a

scheduled first demodulation reference signal, DM-RS in a slot. The position information may be obtained, for example from the wireless device **16** via radio resource control scheduling of the DM-RS. The method also includes one of transmitting and receiving (block **S122**) the PT-RS within the slot, where the position of the PT-RS depends on the position in the time domain of the scheduled first DM-RS.

(57) In some embodiments, the obtaining comprises one of receiving information about and determining the position in the time domain of the scheduled first DM-RS. In some embodiments, the method further includes obtaining information about a position of a first time symbol in a slot scheduled for data transmission. In some embodiments, the method also includes obtaining information about a position of a last time symbol in the slot scheduled for data transmission. In some embodiments, the method also includes obtaining information indicating a scheduled modulation and coding scheme, MCS, and transmitting the PT-RS with a time density based on the scheduled MCS. This may be the case for OFDM, whereas for DFTS-OFDM waveforms, higher layer messaging may be used to indicate the time density. In some embodiments, the time density is one 1, $\frac{1}{2}$ and $\frac{1}{4}$. In some embodiments, the method includes mapping the PT-RS to resource elements, REs, in the slot based on one or more of a position of the scheduled first DM-RS, a scheduled MCS, a required time-density, a position of the first time symbol scheduled for data transmission and a position of the last time symbol scheduled for data transmission. In some embodiments, the first DM-RS is scheduled in resource elements, REs, which span several subcarriers in frequency and one or two time symbols of the slot in time, while the PT-RS is one of transmitted and received in REs which span at least one subcarrier in frequency and multiple time symbols of the slot in time. In some embodiments, a physical resource block, PRB, of the slot has 12 subcarriers in the frequency domain and one of 12 and 14 time symbols in the time domain. In some embodiments, the radio node **14**, **16** is one of a WD and a network node. In some embodiments, an additional DM-RS is scheduled in the same slot as the first DM-RS and PT-RS, and the position of the second DM-RS depends on the position of the PT-RS. In some embodiments, the PT-RS is only transmitted in the mm wavelength high frequency band.

(58) According to another aspect, a radio node **14**, **16** in a wireless communication system configured for one of transmitting and receiving a phase tracking-reference signal, PT-RS, is provided. The radio node **14**, **16** includes processing circuitry configured to obtain information about a position in a time domain of a scheduled first demodulation reference signal, DM-RS in a slot. The processing circuitry **22**, **42** is further configured to one of transmit and receive the PT-RS within the slot, where the position of the PT-RS depends on the position in the time domain of the scheduled first DM-RS.

(59) According to this aspect, in some embodiments, the obtaining comprises one of receiving and determining information about the position in the time domain of the scheduled first DM-RS. In some embodiments, the processing circuitry **22**, **42** is further configured to obtain information about a position of a first time symbol in a slot scheduled for data transmission, and obtain information about a position of a last time symbol in the slot scheduled for data transmission. In some embodiments, the processing circuitry **22**, **42** is further configured to obtain information indicating a scheduled modulation and coding scheme, MCS, and transmit the PT-RS with a time density based on the scheduled MCS. In some embodiments, the time density is one 1, $\frac{1}{2}$ and $\frac{1}{4}$. In some embodiments, the processing circuitry **22**, **42** is further configured to map the PT-RS to resource elements, REs, in the slot based on one or more of a position of the scheduled first DM-RS, a scheduled MCS, a required time-density, a position of the first time symbol scheduled for data transmission and a position of the last time symbol scheduled for data transmission. In some embodiments, the first DM-RS is scheduled in resource elements, REs, which span several subcarriers in frequency and at least one time symbol of the slot in time, while the PT-RS is one of transmitted and received in REs which span at least one subcarrier in frequency and multiple time symbols of the slot in time. In some embodiments, a physical resource block, PRB, of the slot has

12 subcarriers in the frequency domain and one of 12 and 14 time symbols in the time domain. In some embodiments, the radio node **14, 16** is one of a WD and a network node.

(60) According to another aspect, a radio node **14, 16** in a wireless communication system configured for one of transmitting and receiving a phase tracking-reference signal, PT-RS is provided. The radio node **14, 16** includes demodulation reference signal, DM-RS, position obtainer module **19, 59** configured to obtain information about a position in a time domain of a scheduled first demodulation reference signal, DM-RS in a slot. The radio node **14, 16** further includes a PT-RS transceiver module **29, 49** configured to one of transmit and receive the PT-RS within the slot, where the position of the PT-RS depends on the position in the time domain of the scheduled first DM-RS. In some embodiments, the first DM-RS is scheduled in resource elements, REs, which span several subcarriers in frequency and at least one time symbol of the slot in time, while the PT-RS is one of transmitted and received in REs which span at least one subcarrier in frequency and multiple time symbols of the slot in time.

(61) Some embodiments include the following: Embodiment 1. A method for scheduling phase tracking reference signals, PT-RS, jointly with demodulation reference signals, DM-RS, the method comprising: scheduling the DM-RS in a time-frequency resource grid at a plurality of frequencies in a same time slot; and scheduling the PT-RS in the time-frequency grid at a plurality of time slots at a same frequency, the time slots at which the PT-RS are scheduled depending on a position of the DM-RS. Embodiment 2. The method of Embodiment 1, wherein a time-density of the PT-RS depends on a selected modulation and coding scheme. Embodiment 3. The method of any of Embodiments 1 and 2, wherein the time-density is one half. Embodiment 4. The method of any of Embodiments 1-3, wherein the DM-RS is single front loaded. Embodiment 5. The method of any of Embodiments 1-3, wherein the DM-RS is double front loaded. Embodiment 6. The method of any of Embodiments 1-5, wherein the method is performed by a network node. Embodiment 7. The method of any of Embodiments 1-5, wherein the method is performed by a wireless device.

Embodiment 8. A network node configured to schedule phase tracking reference signals, PT-RS, jointly with demodulation reference signals, DM-RS, the network node comprising: processing circuitry configured to: schedule the DM-RS in a time-frequency resource grid at a plurality of frequencies in a same time slot; and schedule the PT-RS in the time-frequency grid at a plurality of time slots at a same frequency, the time slots at which the PT-RS are scheduled depending on a position of the DM-RS. Embodiment 9. The network node of Embodiments 8, wherein a time-density of the PT-RS depends on a selected modulation and coding scheme. Embodiment 10. The network node of any of Embodiments 8 and 9, wherein the time-density is one half. Embodiment 11. The network node of any of Embodiments 8-10, wherein the DM-RS is single front loaded. Embodiment 12. The network node of any of Embodiments 8-10, wherein the DM-RS is double front loaded. Embodiment 13. A network node configured to schedule phase tracking reference signals, PT-RS, jointly with demodulation reference signals, DM-RS, the network node comprising: a DM-RS scheduling module configured to schedule the DM-RS in a time-frequency resource grid at a plurality of frequencies in a same time slot; and a PT-RS scheduling module configured to schedule the PT-RS in the time-frequency grid at a plurality of time slots at a same frequency, the time slots at which the PT-RS are scheduled depending on a position of the DM-RS. Embodiment 14. A wireless device configured to schedule phase tracking reference signals, PT-RS, jointly with demodulation reference signals, DM-RS, the wireless device comprising: processing circuitry configured to: schedule the DM-RS in a time-frequency resource grid at a plurality of frequencies in a same time slot; and schedule the PT-RS in the time-frequency grid at a plurality of time slots at a same frequency, the time slots at which the PT-RS are scheduled depending on a position of the DM-RS. Embodiment 15. The network node of claim **14**, wherein a time-density of the PT-RS depends on a selected modulation and coding scheme. Embodiment 16. The network node of any of Embodiments 14 and 15, wherein the time-density is one half. Embodiment 17. The network node of any of Embodiments 14-16, wherein the DM-RS is single front loaded. Embodiment 18.

The network node of any of Embodiments 14-16, wherein the DM-RS is double front loaded.
Embodiment 19. A wireless device configured to schedule phase tracking reference signals, PT-RS, jointly with demodulation reference signals, DM-RS, the wireless device comprising: a DM-RS scheduling module configured to schedule the DM-RS in a time-frequency resource grid at a plurality of frequencies in a same time slot; and a PT-RS scheduling module configured to schedule the PT-RS in the time-frequency grid at a plurality of time slots at a same frequency, the time slots at which the PT-RS are scheduled depending on a position of the DM-RS.

(62) TABLE-US-00001 Abbreviation Explanation 3GPP Third Generation Partnership Project eNB Enhanced NodeB CRS Cell-Specific Reference Signal DM-RS De-Modulation Reference Signal DCI Downlink Control Information LTE Long Term Evolution MIMO Multiple Input Multiple Output PT-RS Phase Tracking Reference Signal RS Reference Signal TM Transmission Mode TTI Transmission Time Interval UE User Equipment URLLC Ultra-reliable low latency communications WD Wireless Device

(63) As will be appreciated by one of skill in the art, the concepts described herein may be embodied as a method, data processing system, and/or computer program product. Accordingly, the concepts described herein may take the form of an entirely hardware embodiment, an entirely software embodiment or an embodiment combining software and hardware aspects all generally referred to herein as a “circuit” or “module.” Furthermore, the disclosure may take the form of a computer program product on a tangible computer usable storage medium having computer program code embodied in the medium that can be executed by a computer. Any suitable tangible computer readable medium may be utilized including hard disks, CD-ROMs, electronic storage devices, optical storage devices, or magnetic storage devices.

(64) Some embodiments are described herein with reference to flowchart illustrations and/or block diagrams of methods, systems and computer program products. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer (to create a special purpose computer), special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

(65) These computer program instructions may also be stored in a computer readable memory or storage medium that can direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer readable memory produce an article of manufacture including instruction means which implement the function/act specified in the flowchart and/or block diagram block or blocks.

(66) The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide steps for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks. It is to be understood that the functions/acts noted in the blocks may occur out of the order noted in the operational illustrations. For example, two blocks shown in succession may in fact be executed substantially concurrently or the blocks may sometimes be executed in the reverse order, depending upon the functionality/acts involved. Although some of the diagrams include arrows on communication paths to show a primary direction of communication, it is to be understood that communication may occur in the opposite direction to the depicted arrows.

(67) Computer program code for carrying out operations of the concepts described herein may be written in an object oriented programming language such as Java® or C++. However, the computer

program code for carrying out operations of the disclosure may also be written in conventional procedural programming languages, such as the “C” programming language. The program code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer. In the latter scenario, the remote computer may be connected to the user's computer through a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

(68) Many different embodiments have been disclosed herein, in connection with the above description and the drawings. It will be understood that it would be unduly repetitious and obfuscating to literally describe and illustrate every combination and subcombination of these embodiments. Accordingly, all embodiments can be combined in any way and/or combination, and the present specification, including the drawings, shall be construed to constitute a complete written description of all combinations and subcombinations of the embodiments described herein, and of the manner and process of making and using them, and shall support claims to any such combination or subcombination.

(69) It will be appreciated by persons skilled in the art that the present embodiments are not limited to what has been particularly shown and described herein above. In addition, unless mention was made above to the contrary, it should be noted that all of the accompanying drawings are not to scale. A variety of modifications and variations are possible in light of the above teachings without departing from the scope of the following claims.

Claims

1. A method for use in a wireless device in a wireless communication system for transmitting a phase-tracking reference signal, PT-RS, the method comprising: obtaining information about a position in a time domain of a scheduled first demodulation reference signal, DM-RS, symbol in a slot; obtaining information about a position of a first time symbol in the slot scheduled for data transmission; obtaining information about a position of a last time symbol in the slot scheduled for data transmission; obtaining information about the PT-RS density; mapping of PT-RS symbols to resource elements, REs, within the slot based on an equation determined by the obtained information about the position in the time domain of the scheduled first DM-RS symbol in the slot, the position of the first time symbol in the slot scheduled for data transmission, the position of the last time symbol in the slot scheduled for data transmission, and the PT-RS density; and transmitting PT-RS symbols within the slot.
2. The method of claim 1, wherein the obtaining comprises one of receiving information about and determining the position in the time domain of the scheduled first DM-RS symbol.
3. The method of claim 1, further comprising: obtaining information indicating a scheduled modulation and coding scheme, MCS; and transmitting the PT-RS with a time density based on the scheduled MCS.
4. The method of claim 3, wherein the time density is one of 1, $\frac{1}{2}$ and $\frac{1}{4}$.
5. The method of claim 1, further comprising: mapping the PT-RS to resource elements, REs, in the slot based on, a scheduled MCS.
6. The method of claim 1, wherein the first scheduled DM-RS symbol is scheduled in resource elements, REs, which span a plurality of subcarriers in frequency and at least one time symbol of the slot in time, while the PT-RS is one of transmitted and received in REs which span at least one subcarrier in frequency and multiple time symbols of the slot in time.
7. The method of claim 1, wherein a physical resource block, PRB, of the slot has 12 subcarriers in a frequency domain and one of 12 and 14 time symbols in the time domain.
8. A wireless device in a wireless communication system configured for one of transmitting and

receiving a phase tracking-reference signal, PT-RS, the wireless device comprising: processing circuitry configured to: obtain information about a position in a time domain of a scheduled first demodulation reference signal, DM-RS, symbol in a slot; obtain information about a position of a first time symbol in a slot scheduled for data transmission; obtain information about a position of a last time symbol in the slot scheduled for data transmission; obtain information about the PT-RS density; and, map PT-RS symbols to resource elements, REs, within the slot based on an equation determined by on the obtained information about the position in the time domain of the scheduled first DM-RS symbol in the slot, the position of the first time symbol in the slot scheduled for data transmission, the position of the last time symbol in the slot scheduled for data transmission, and the PT-RS density; and transmit PT-RS symbols within the slot.

9. The wireless device of claim 8, wherein the obtaining comprises one of receiving and determining information about the position in the time domain of the scheduled first DM-RS symbol.

10. The wireless device of claim 8, wherein the processing circuitry is further configured to: obtain information indicating a scheduled modulation and coding scheme, MCS; and transmit the PT-RS with a time density based on the scheduled MCS.

11. The wireless device of claim 10, wherein the time density is one of 1, $\frac{1}{2}$ and $\frac{1}{4}$.

12. The wireless device of claim 8, wherein the processing circuitry is further configured to: map the PT-RS to resource elements, REs, in the slot based on, a scheduled MCS.

13. The wireless device of claim 8, wherein the scheduled first DM-RS symbol is scheduled in resource elements, REs, which span a plurality of subcarriers in frequency and at least one time symbol of the slot in time, while the PT-RS is one of transmitted and received in REs which span at least one subcarrier in frequency and multiple time symbols of the slot in time.

14. The wireless device of claim 8, wherein a physical resource block, PRB, of the slot has 12 subcarriers in a frequency domain and one of 12 and 14 time symbols in the time domain.
